Jehan Birdy

Los Angeles, United States of America | +1 2139947284 | jehan.birdy@gmail.com | Design Portfolio

EDUCATION

University of Southern California **Aug 2025 - Present**
Master of Science, Aerospace Engineering *Los Angeles, U.S.A.*
• **GPA:** 3.35/4.0

K.J. Somaiya College of Engineering **Jul 2021 - May 2025**
Bachelor of Technology, Mechanical Engineering *Mumbai, India*
• **GPA:** 8.38/10

SKILLS

• **Design Software**: Siemens NX, SolidWorks, Fusion 360, CATIA V5.
• **Simulation & Engineering**: ANSYS Workbench, ANSYS ACP, Simcenter STAR-CCM+, Simulink, MATLAB.
• **Programming & Development Tools**: ROS2, Python, Git, Linux, MuJoCo.

EXPERIENCE

Tata Lockheed Martin Aerostructures Limited **Jun 2025 - Aug 2025**
Manufacturing Intern *Hyderabad, India*
• Supported the planned progress of the Airbus C295 floor panel program's metal-to-metal bonding operations, helping prevent manufacturing delays. Worked alongside concurrent OEM programs, including the Lockheed Martin C-130J and F-16 and Boeing AH-64 Apache
• Developed a consumables estimation sheet for C295 aluminum honeycomb sandwich panel components and used CATIA V5 to verify hole dimensions and compute surface areas for primer and paint estimation applying 30% wastage.
• Corrected Manufacturing Instruction Sheets on the factory floor and generated all drilling tool orders for the C295 FAL (Final Assembly Line) components. I raised multiple Engineering Query Numbers (EQNs) to Airbus on dimensional and slot-specifications and updated the SAP system with process guideline changes and cross-checked BOM errors.
• Observed end-to-end manufacture of C295 sandwich panels, with PAA anodizing, clean room film adhesive layup, autoclave curing, and post-cure edge vacuum sealing - and the full C-130J metal-to-metal bonding process including ultrasonic NDT and destructive shear witness coupon testing.

Careflex Private Limited **Jun 2025 - Jul 2025**
Manufacturing Intern *Mumbai, India*
• Designed and implemented various systems and tools to improve the manufacturing and quality control of reusable casts for wrist fractures. This included a setup for efficient cast testing to detect deformations or defects, and molds for silicone pouring at connector sites
• Manufactured a 2D CNC machine using a combination of aluminum extrusion tubes and 3D-printed components to embed heating elements, which helped achieve manufacturing self-sufficiency

BETIC Lab of IIT Bombay **Jan 2025 - May 2025**
Product Development Intern *Mumbai, India*
• I worked alongside teams of engineers on the development of multiple projects such as the Scalp Cooling Cap, Diabetic Foot Sensor and Tracheostomy Tube, each designed to tackle varied medical conditions.
• Worked on medical-device development by engaging with biomedical engineers and clinicians on user-centric design, generating design concepts, creating 3D CAD models in SOLIDWORKS, and developing functional prototypes and components through 3D printing and CNC machining
• I conducted testing for all projects, including the Scalp Cooling Cap, to ensure that no errors would arise during operation, and used Figma to design its user-interface screens
• Helped create a functioning prototype of the Scalp Cooling Cap by the end of the internship and presented designs for the surgical Tracheostomy Tube

Bajaj Auto **Jun 2024 - Jun 2024**
Mechanical Intern *Pune, India*
• Collaborated with the Manufacturing Engineering team at Chakan Plant-1 to analyze assembly processes and apply error-proofing (Poka-Yoke) tools to minimize defects, rework, and waste while improving production efficiency
• Created a comprehensive report on the current assembly-line layout and existing error-proofing tools, including recommendations for additional tools where necessary and conveyor-belt interlocking to prevent human error

EXTRACURRICULAR

Advanced Composites Design Lab, USC **Jan 2026 - Present**
Sub-team Lead *Los Angeles, USA*

- As a lead of Professor Bo Jin's Lab, I design and manufacturing of glass fiber beams for the Society for the Advancement of Material and Process Engineering (SAMPE) Bridge International Competition. I learnt and used ANSYS ACP to create composite glass fiber square beam simulations to submit successful designs to SAMPE 2025 & 2026.
- I worked on finding the strongest layup designs to ensure minimum weight without compromising on strength, while also refining our manufacturing processes to remove any potential error points.
- I submitted designs which were considered for the 2026 SAMPE competition was held at the University of Washington, Seattle, where we secured 3rd place in our category as well as 3rd place in the overall standings. I will also be taking part in the 2027 SAMPE competition.

Formula Student Team, USC Racing Team **Sep 2025 - Present**
Designer *Los Angeles, USA*

- As a member of the aerodynamics team of USC Racing, I designed front wing whiskers using Siemens NX to limit the wind turbulence from the front wing on the rear wing and learnt about the use of STAR-CCM+ for Computational Fluid Dynamics.
- Designed and manufactured the molds for the manufacturing of carbon fiber aerodynamic components such as the Front Wing, Rear Wing, Diffuser and Undertray. Experimented with molds of varying density in order to achieve the optimum surface finish.
- I was also a part of the aerodynamic component manufacturing team, where I helped in the manufacturing of our Front Wing, Rear Wing, Undertray as well as other smaller components, using processes such as 3D printing and Carbon Fiber layups with Vacuum infusion.
- At the Formula SAE 2026 held in Michigan, out of the 131 participating university teams, USC Racing secured 1st position in the Autocross event, 5th in Design, 7th in Endurance and 7th position overall.

Formula Student Team, Orion Racing India **Jan 2022 - Jun 2023**
Designer *Mumbai, India*

- Designed and manufactured vehicle chassis for multiple domestic and international Formula Student competitions, learning the processes for the construction of a Formula 1 style single-seater car.
- Coordinated with vendors to source materials including AISI 4130 tubes for the space frame chassis and battery cells for the battery pack, and managed manufacturing processes such as cutting, spot-welding, and anodizing chassis.
- Used SolidWorks and Ansys to design and check compliance of the chassis with competition rules.

ACADEMIC PROJECTS'

- Built a solar-powered mobile charger as part of a project focused on sustainability.
- Collaborated with a team of 4 to design and 3D print a battery-operated snake robot for surveying pipes and accessing tight spaces.
- Modelled and simulated an autopilot stabilization system for an unpowered glider using MATLAB and Simulink, enabling stable flight from varying launch conditions and autonomous tracking of a predetermined flight path through landing.